

KASHISH

📍 Sukkur, Sindh, Pakistan | 📞 +92 337 2068803 | ✉ kashishbasreja.fss22@gmail.com | 🔗 LinkedIn | 🐙 GitHub | 🌐 Portfolio

Professional Summary

Computer Science graduate (CGPA: 3.66 upto 7th Semester) specializing in AI/ML and full-stack software development. Hands-on experience in deep learning, computer vision, NLP, and LLM fine-tuning; research accepted at iCoMET 2026. Seeking entry-level roles in AI, Machine Learning, and Software Engineering.

Education

Sukkur IBA University

B.S. Computer Science **CGPA: 3.66 / 4.00**

Sep 2022 – May 2026

Sukkur, Pakistan

- Coursework: Data Structures, Algorithms, Deep Learning, Natural Language Processing, Computer Vision, Web Engineering, Object Oriented Programming, Machine Learning.
- **STHP - Sindh Talent Hunt Program**; Fully Funded 4-year merit based scholarship

Experience

Web Development Intern

Sukkur Electric Power Company (SEPCO)

Jun 2025 – Jul 2025

Sukkur, Pakistan

- Developed responsive web pages; collaborated with cross-functional teams to implement features, optimize performance, and debug issues.

WordPress Developer Intern

Students Empowered Oasis (SEO)

Jul 2024 – Sep 2024

Remote

- Built and managed secure WordPress sites; customized enterprise themes, custom plugins, and responsive layouts for seamless user experiences.

Research & Publications

“Reducing Unsafe Mental Health Advice in LLMs Using Safety-Aware Prompting”

Accepted – iCoMET 2026

- Evaluated multiple instruction-tuned LLMs on a real-world mental health counseling dataset using a zero-shot safety-aware prompting strategy, achieving complete elimination of dangerous divergence across all models without any fine-tuning.
- Introduced the **dangerous divergence** metric to jointly quantify unsafe and semantically misaligned LLM responses; demonstrating that safety-aware prompting reduces unsafe rates by up to 100% while preserving semantic fidelity to human therapist responses.

Projects

NeuroEmotion – Final Year Project | EEG, Deep Learning, Flutter, Flask

May 2026

- Affective computing platform for real-time multi-user emotion classification from raw EEG brainwaves and facial mapping, integrating signal preprocessing pipelines with deep learning inference served via a Flask backend.
- Designed a cross-platform Flutter frontend to visualize emotion states in real time, enabling accessible brain-computer interface applications targeted at mental wellness and human-computer interaction research.

Ocular Disease Detection | PyTorch, ResNet50, VGG16, EfficientNet-B3

Apr 2026

- Built a multi-label CNN classification pipeline benchmarking three architectures (ResNet50, VGG16, EfficientNet-B3) on the ODIR fundus dataset to simultaneously screen retinal samples for 8 distinct pathologies including glaucoma, diabetic retinopathy, and macular degeneration.
- Implemented comprehensive evaluation with per-class AUC, F1, and confusion matrix analysis, providing a reproducible diagnostic benchmark for clinical AI research in ophthalmology.

AI Sign Language Interpreter | OpenCV, MediaPipe, CNN, gTTS

- Real-time gesture recognition system supporting ASL, BSL, and ISL using five custom-trained CNN topologies on hand landmark features extracted via MediaPipe, achieving 96% classification accuracy with low-latency audio feedback via gTTS.
- Designed the pipeline to operate on standard webcam hardware with no GPU requirement, making the assistive tool accessible to users with speech and hearing impairments.

Healr | MongoDB, Express.js, React, Node.js

Academic

- Full-stack MERN health management platform featuring secure role-based access control for three user roles; Patients, Doctors, and Admins, with dedicated dashboards for appointment scheduling, medical records, and administrative oversight.
- Implemented JWT-based authentication, protected API routing, and a responsive React frontend with modular component architecture, following industry-standard practices for scalable web application development.

Volunteering

Web Developer (Volunteer) – Students Empowered Oasis

Sep 2024 – Apr 2026

- Administered content platforms, engineered automated quiz matrices, published monthly technical newsletters, and deployed responsive event systems.

Technical Skills

Languages: Python, C++, Java, JavaScript, SQL, HTML/CSS

ML / DL Frameworks: PyTorch, TensorFlow, Keras, Scikit-Learn, OpenCV, MediaPipe, Hugging Face, Transformers

Domains: Computer Vision (detection, classification, segmentation, tracking), NLP, LLM Fine-tuning (LoRA), EEG Signal Processing

MLOps & Tools: Docker, Linux, Git/GitHub, Jupyter, Google Colab, Kaggle, Ollama, LangChain, VS Code

Backend / Web: Node.js, Express.js, Flask, React, MongoDB, WordPress

Honors, Awards & Achievements

- **Best Web Developer of the Year:** Students Empowered Oasis, awarded for dedicated platform management and volunteer contributions as a regular web developer.
- **SIBAFest Speed Competition Winner:** First place in competitive speed programming event at Sukkur IBA University.
- **PM Youth Laptop Scheme 2023:** Awarded for academic excellence (CGPA 3.79).
- **Certifications:** Google Prompting Essentials (Coursera) | AI for Everyone (DeepLearning.AI) | C++ Competition 1st Place
- **Leadership:** Executive Member, CS Society (2025–2026) | Active Member, SIBA Tech Club (2024–2025)